

YASH KUMAR VERMA

AI ENGINEER | GenAI & Full-Stack Developer

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[Portfolio](#) | Available for Full-Time Roles Immediately | Open to Relocation

PROFESSIONAL SUMMARY

AI Engineer (B.Tech CSE, GPA 7.87, 2026) specializing in **LLM, RAG, and multi-agent systems** — from pipeline design to deployment. Built production-grade agentic architectures using **LangChain/LangGraph, vector databases, and quantitative evaluation frameworks (Ragas)**. Strong grounding in **Python, backend systems, and full-stack integration** for shipping AI features end-to-end.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript (ES2022+), TypeScript
- **AI/LLM Engineering:** LangChain, LangGraph, RAG, Prompt Engineering, Multi-Agent Systems, OpenAI API, Gemini API, LLM Evaluation (Ragas), Vector Databases (ChromaDB, Pinecone, Qdrant), Cross-Encoder Reranking, Embeddings
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Pandas, NumPy, CNNs, Model Fine-Tuning
- **Backend & APIs:** Node.js, Express, FastAPI, REST APIs, Microservices, JWT Auth, Redis, MySQL, MongoDB, PostgreSQL/PostGIS
- **Frontend & Mobile:** React, Next.js, React Native, Tailwind CSS, Vite
- **DevOps & Tools:** Docker, Git, AWS (EC2/Lambda), GCP, CI/CD, Postman, Vercel, Firebase

AI PROJECTS

HyperGraph-CDC: Real-Time Event-Driven Knowledge Graph & MCP Agent Gateway | [GitHub](#)

PostgreSQL, Debezium, Kafka, Neo4j, Qdrant, LangGraph, MCP — 2026

- Built a real-time, event-driven data ingestion pipeline achieving <15ms database capture latency and a 1.1s end-to-end graph indexing turnaround, by capturing PostgreSQL WAL changes via Debezium, streaming through Apache Kafka, and writing asynchronously to Neo4j and Qdrant.
- Developed a hybrid GraphRAG search engine that reduced grounding context blind spots while maintaining retrieval speeds under 69ms, by extracting entity/relationship structure with Gemini 2.5 Flash and combining dense vector search with Cypher-based k-hop graph traversals.
- Architected a decoupled LLM supervisor agent gateway, reducing agent-to-tool integration overhead to <13ms, by implementing the Model Context Protocol (MCP) over Stdio transport to dynamically discover and route queries across graph, vector, web search, and codebase-indexing tools with a LangGraph router.
- Benchmarked the full stack end-to-end, validating sub-5ms Qdrant/Neo4j retrieval at 100K+ vectors and throughput up to 25 concurrent write operations/sec bound by LLM extraction concurrency.
- Exposed full-stack observability through a FastAPI supervisor gateway and a React/Vite live dashboard, surfacing real-time MCP tool-execution traces, server connectivity status, and CDC pipeline health.

Cortex AI — Multi-Agent AI Platform

Node.js, React, LangGraph.js, Redis, MongoDB, Qdrant, Microservices — 2026

- Architected a production-grade multi-agent AI platform on a microservices backend (Auth, Chat, Agent, Billing services behind an API Gateway), orchestrating 8 specialized agents via a LangGraph.js supervisor that auto-classifies intent and routes requests.
- Built specialized agents for code generation (DeepSeek), web search (Tavily), PDF/PPT generation, AI image generation (Pollinations.ai), and multimodal vision (Gemini), streaming structured artifacts to a React 19 + Monaco Editor frontend.
- Implemented a PDF RAG agent using Gemini embeddings and Qdrant Cloud (chunking, similarity search, per-query collection cleanup) plus Redis-backed conversation memory (20-message sliding window) and Firebase session auth.
- Designed the API Gateway with Firebase Admin SDK auth, Redis session store, Helmet security headers, and per-agent rate limiting; integrated Backblaze B2 (S3-compatible) storage for generated files.
- Instrumented full-platform observability with LangSmith tracing, structured MongoDB execution traces, and Redis metric aggregation, paired with an LLM-as-a-judge evaluation pipeline auto-scoring responses on faithfulness, relevance, and safety.
- Built a Redis-backed semantic cache using cosine-similarity matching to cut duplicate-query latency and LLM API cost across the agent platform.

Local Talk-to-Codebase RAG (CodeLens) | [GitHub](#)

LangChain, LangGraph, ChromaDB, Gemini, Ragas, Cross-Encoder Reranking — 2026

- Built a fully local retrieval-augmented generation system enabling natural-language Q&A over any codebase — combining LangGraph orchestration, ChromaDB vector storage, and Gemini for grounded, code-aware responses.
- Implemented Cross-Encoder reranking to sharpen retrieval precision and built an automated evaluation harness with Ragas (context precision, recall, faithfulness, answer relevancy) to continuously benchmark pipeline quality, reaching an internally validated 8/10 output-quality score.

- Debugged and hardened the pipeline in production-style conditions — resolving silent failure modes, stale reranker references, and architecting multi-tenant session isolation for concurrent codebase queries.
- Shipped a production-ready deployment: multi-stage Dockerized build with persistent ChromaDB volumes and a GitHub Actions CI/CD pipeline that runs automated Ragas regression tests on every push before building the release image.

Agent Observability Platform | [GitHub](#)

Python, FastAPI, PostgreSQL, React, Recharts, Docker, LangChain/LangGraph — 2026

- Built a plug-and-play Python instrumentation SDK for LangChain/LangGraph agents that streams trace data to a FastAPI collector with zero blocking overhead, guaranteeing complete telemetry delivery via async task tracking that awaits full step-flush before connection teardown.
- Engineered a rule-based failure classification engine that detects timeouts (>30s), infinite loops, and hallucinations directly from trace content — requiring zero external LLM calls — while background workers enforce try/finally DB session lifecycle guarantees to eliminate connection leaks.
- Designed a cross-version regression detection system that raises alerts when success rates drop >20% between deployments, using chronological version-timestamp ordering instead of naive alphabetical sort to correctly sequence arbitrary version tags (e.g., v2 vs. v10).
- Shipped a real-time React/Recharts dashboard with 3-second auto-refresh polling, dual-version latency and success-rate comparative analytics, and per-run cost tracking from token usage, paired with a LangGraph-based evaluation harness that simulates healthy vs. degraded agent versions to generate demo regression data.

EXPERIENCE

Tellis Technologies — Software Development Intern

MERN Stack, Agentic AI, LangChain/LangGraph, Docker, Tree-sitter — 2025

- Built agentic AI pipelines using LangChain and LangGraph, integrating LLM-driven orchestration into production MERN applications alongside the engineering team.
- Built a production-grade multi-agent system (Autonomous SWE Agent) that autonomously localizes bugs, writes patches, and iterates on real GitHub issues, benchmarked end-to-end against SWE-bench-lite.
- Designed a LangGraph self-verification loop (Planner → Editor → Test Runner → Verifier) applying patches inside an isolated Docker sandbox, running the real test suite, and routing failures back to the Editor with error traces for iterative retries.
- Implemented AST-level semantic retrieval using tree-sitter to extract full functions/classes (not naive text chunks), embedded via ChromaDB with Gemini embeddings for accurate issue-to-code matching.

Elite Coders — Full Stack Development Intern

MERN Stack, School ERP, Database Optimization — 2025

- Developed core modules of a School ERP system (student records, attendance, fee management) using the MERN stack for real institutional use.
- Optimized MongoDB schema design and indexing on high-traffic collections, improving query performance for the ERP's reporting workflows.
- Built and secured RESTful APIs with role-based access control (admin, teacher, student) for multi-tenant institutional access.

EDUCATION

B.Tech in Computer Science & Engineering | Sep 2022 – Jun 2026

Vishveshwarya Group of Institutions, Greater Noida, UP | GPA: 7.87 / 10

LEADERSHIP & ACHIEVEMENTS

- **Hackathons:** Top-3 finishes at 8+ inter-college hackathons, building and shipping functional AI/full-stack prototypes under time constraints.
- **Research:** Co-authored a peer-reviewed academic paper on agentic RAG and authored a 60+ page technical report for a Ministry of Earth Sciences-sponsored project (SIH-25040/INCOIS).